Definition: An equation which involving the differential coefficient is called differential equation.

(i) $\frac{dy}{dx}+y=0$ (ii) $\frac{\partial^2 z}{\partial x^2}-a^2\frac{\partial^2 z}{\partial y^2}=0$

Ordinary differential equation: An ordinary differential equation is one which contains differential coefficient of a single independent variable.

$$\frac{dy}{dx}+2y=0$$

Partial differentiation equation: A partial differential equation is one which contains partial differential co-efficient and as such involves two or more independent variables.

* The order of a differential equation is the order of the highest derivatives.

* The degree of differential equation is the degree of the highest order in the differential equation.

Any relation connecting the variables of an equation are not involving their derivatives, which satisfies the given differential equation and also the given differential equation can be derived is called a solution of the differential equations.

* The solution of a differential equation in which the number of arbitrary constants is equal to the order of the differential equation is called general or complete solution of differential equation.

* The solutions obtained by given particular values ~~of~~ to the arbitrary constants of the general solution is called aparticular solution of the differential equation.

Differential equation form করতে হলে যতগুলো constant থাকে ততবার derivative করে constant কে omit করতে হবে।

Ex If the centre of a circle lies in the x-axis. form the differential . which ~~centre~~ pass~~ing~~es through the origin.

Soln: let $(a,0)$ be the centre and a be the radius of the circle.

Then the equation of circle,

$$(x-a)^2+y^2=a^2$$

$$\Rightarrow x^2+y^2-2ax=0 \quad \text{---- (1)}$$

differentiate eqn (1) with respect to x,

$$2x+2y\cdot\frac{dy}{dx}=2a$$

$$\Rightarrow 2a=2x+2y\cdot\frac{dy}{dx}$$

Substituing the value in eqn (1)

$$x^2+y^2-\cancel{x}(2x+2y\frac{dy}{dx})=0$$

$$\Rightarrow y^2-x^2-2yx\frac{dy}{dx}=0.$$

(Ans)

日 Form the differential eqⁿ for $r = a(1+\cos\theta)$

Solⁿ: Here, $r = a(1+\cos\theta)$ - - - - ①

differentiate eqⁿ ① with respect to θ

$$\frac{dr}{d\theta} = -a\sin\theta$$

$$\therefore a = -\frac{1}{\sin\theta}\cdot\frac{dr}{d\theta} \text{ - - - - (ii)}$$

Combining ① and (ii) we get,

$$r = -\frac{1}{\sin\theta}(1+\cos\theta)\frac{dr}{d\theta}$$

$$\Rightarrow r\sin\theta = -(1+\cos\theta)\frac{dr}{d\theta}$$

$$\therefore (1+\cos\theta)\frac{dr}{d\theta} + r\sin\theta = 0.$$

(Ans)

* $y = y(x) = A\cos x + B\sin x$

Solⁿ: Here, $y = A\cos x + B\sin x$ - - - - ①

differentiate eqⁿ ① with respect to x,

$$y_1 = -A\sin x + B\cos x$$

$$\Rightarrow y_2 = -A\cos x - B\sin x$$

$$= -(A\cos x + B\sin x)$$

$$= -y$$

$$\therefore y_2 = -y$$

that means, $\frac{d^2y}{dx^2} + y = 0$ (Ans)

$y = e^{x} (A\cos x + B\sin x)$

<u>Soln:</u> Here, $y = e^{x} (A\cos x + B\sin x)$ ---- (1)

differentiate eqn (1) with respect to x

$y_1 = e^{x}(A\cos x + B\sin x) + e^{x}(-A\sin x + B\cos x)$

$\Rightarrow y_1 = y + e^{x}(-A\sin x + B\cos x)$

$\Rightarrow y_2 = y_1 + e^{x}(-A\sin x + B\cos x) + e^{x}(-A\cos x - B\sin x)$

$\Rightarrow y_2 = y_1 + (y_1 - y) - y$ problm.

$\Rightarrow y_2 = y_1 + y_1 - y - y$

$\Rightarrow y_2 = 2y_1 - 2y$

$\Rightarrow y_2 - 2y_1 + 2y = 0.$

Ans

$$y_1 = e^{x}(\cancel{A\cos x + B\sin x}) + e^{x}(-A\sin x + B\cos x)$$
$$y = e^{x}(\cancel{A\cos x + B\sin x})$$
$$y_1 - y = e^{x}(-A\sin x + B\cos x)$$

$\boxed{\begin{matrix} y \\ y_1 \\ y_2 \\ y_3 \\ y_4 \end{matrix}}$ y_5 =

$\boxed{\begin{matrix} y \\ y_1 \\ y_2 \end{matrix}}$ y_3.

Seperation of variable

(1) A differential equation of the form $f(x)dx + f(y)dy = 0$

(1) Solve the differential equation $x\sqrt{1-y^2}\,dx + y\sqrt{1-x^2}\,dy = 0$

Soln: Given that,

$$x\sqrt{1-y^2}\,dx + y\sqrt{1-x^2}\,dy = 0$$

$$\Rightarrow \frac{x}{\sqrt{1-x^2}}dx + \frac{y}{\sqrt{1-y^2}}dy = 0$$

$$\Rightarrow -\frac{1}{2}2\sqrt{1-x^2} - \frac{1}{2}\cdot 2\sqrt{1-y^2} = -c \quad [\text{by integrating}]$$

$$\Rightarrow \sqrt{1-x^2} + \sqrt{1-y^2} = c. \quad \text{(Ans)}$$

(ii) $y\,dx - x\,dy = xy\,dx$; solve the diff. eqn.

Soln: Given that,

$$y\,dx - x\,dy = xy\,dx$$

$$\Rightarrow \frac{1}{x}dx - \frac{1}{y}dy = dx$$

$$\Rightarrow \left(\frac{1}{x} - 1\right)dx - \frac{1}{y}dy = 0$$

by integrating we get,

$$\ln x - x - \ln y = c$$

$$\Rightarrow \ln\frac{x}{y} = x + c$$

$$\Rightarrow \frac{x}{y} = \cancel{e^{x+c}}\ e^{x+c}$$

$$\Rightarrow \frac{x}{y} = e^x \cdot e^c = e^x c' \quad [\because e^c = c']$$

(Ans)

problem

$$\Rightarrow -\frac{1}{y}dy = -(1/x - 1)dx$$

$$\Rightarrow \frac{1}{y}dy = \frac{1}{x}dx + 1dx$$

$$\Rightarrow \ln y = \ln x - x + c$$

$$\Rightarrow \ln y - \ln x = -x + c$$

$$\Rightarrow \ln y/x = -x + c$$

$$\Rightarrow y/x = e^{-x+c} = e^{-x}e^{c}$$

$$y/x = e^{-x}c'$$

(iii) Solve the equation: $\frac{dy}{dx} = (4x+y+1)^2$ — (1)

Sol^n: let, $4x+y+1 = z$

$$\Rightarrow 4 + \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 4$$

from eq^n (1)

$$\frac{dz}{dx} - 4 = z^2$$

$$\Rightarrow \frac{dz}{dx} = z^2 + 4$$

$$\Rightarrow \frac{dz}{z^2+4} = dx$$

by integrating we get,

$$\frac{1}{2}\tan^{-1}\frac{z}{2} = x + \frac{c}{2}$$

integral →

$$\therefore \tan^{-1}\frac{(4x+y+1)}{2} = 2x + c.$$

(Ans)

(iv) $\frac{dy}{dx} = \frac{y^2+y+1}{x^2+x+1}$

$$\Rightarrow \frac{dy}{y^2+y+1} = \frac{dx}{x^2+x+1}$$

(iv) $\sin^{-1}\left(\frac{dy}{dx}\right) = x+y$

$\Rightarrow \frac{dy}{dx} = \sin(x+y) \quad \text{--- (1)}$

Let, $x+y = z$

$1 + \frac{dy}{dx} = \frac{dz}{dx}$

$\Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 1$

$(1) \Rightarrow \frac{dz}{dx} - 1 = \sin z$

$\Rightarrow \frac{dz}{dx} = \sin z + 1$

$\Rightarrow \frac{dz}{dx} = 1 + \sin z$

$\Rightarrow \frac{dz}{1+\sin z} = dx$

$\Rightarrow \int \frac{dz}{1+\sin z} = \int dx$

$\Rightarrow \int \frac{1-\sin z}{1-\sin^2 z}\, dz = \int dx$

$\Rightarrow \int \frac{1-\sin z}{\cos^2 z}\, dz = \int dx$

$\Rightarrow \int (\sec^2 z - \tan z \sec z)\, dz = \int dx$

$\Rightarrow \tan z - \sec z = x + c$

$\therefore \tan(x+y) - \sec(x+y) = x + c.$

(Ans)

(v) $dy = (y^2 - 1)\,dx$

$\Rightarrow \dfrac{dy}{y^2-1} = dx$

$\Rightarrow \displaystyle\int \dfrac{dy}{y^2-1} = \int dx$

$\Rightarrow \dfrac{1}{2} \log \dfrac{y-1}{y+1} = x + \dfrac{c}{2}$

$\Rightarrow \log \dfrac{y-1}{y+1} = 2x + c.$ (Ans)

(vi) $x \ln x\, dy + \sqrt{1+y^2}\, dx = 0$

$\Rightarrow \dfrac{dy}{\sqrt{1+y^2}} + \dfrac{dx}{x \ln x} = 0$

$\Rightarrow \ln(y + \sqrt{1+y^2}) + \log(\ln x) = \log c$

$\Rightarrow (y + \sqrt{1+y^2}) \log x = c.$ (Ans)

(2ii) $(xy^2 + x)\,dx + (yx^2 + y)\,dy = 0$

$\Rightarrow x(y^2+1)\,dx + y(x^2+1)\,dy = 0$

$\Rightarrow \dfrac{x\,dx}{x^2+1} + \dfrac{y\,dy}{y^2+1} = 0$

$\Rightarrow \dfrac{1}{2}\log(x^2+1) + \dfrac{1}{2}\log(y^2+1) = \dfrac{1}{2}\log c$

$\Rightarrow \log(x^2+1) + \log(y^2+1) = \log c$

$\Rightarrow \log (x^2+1)(y^2+1) = \log c$

$\Rightarrow (x^2+1)(y^2+1) = c.$ (Ans)

(vi) $y\sqrt{x^2-1}\,dx + x\sqrt{y^2-1}\,dy = 0$

$\Rightarrow \frac{\sqrt{x^2-1}}{x}\,dx + \frac{\sqrt{y^2-1}}{y}\,dy = 0$

$\Rightarrow \frac{x^2-1}{x\sqrt{x^2-1}}\,dx + \frac{y^2-1}{y\sqrt{y^2-1}}\,dy = 0$

$\Rightarrow \left(\frac{x}{\sqrt{x^2-1}} - \frac{1}{x\sqrt{x^2-1}}\right)dx + \left(\frac{y}{\sqrt{y^2-1}} - \frac{1}{y\sqrt{y^2-1}}\right)dy = 0$

$\Rightarrow \frac{1}{2}\,2\sqrt{x^2-1} - \sec^{-1}x + \frac{1}{2}\,2\sqrt{y^2-1} - \sec^{-1}y = c$

$\Rightarrow \sqrt{x^2-1} - \sec^{-1}x + \sqrt{y^2-1} - \sec^{-1}y = c.$

(Ans)

* $x\cos^2y\,dx - y\cos^2x\,dy = 0$

$\Rightarrow \frac{x}{\cos^2x}\,dx - \frac{y}{\cos^2y}\,dy = 0$

$\Rightarrow x\sec^2x\,dx - y\sec^2y\,dy = 0$

$\Rightarrow (x\tan x - \log\sec x) - (y\tan y - \log\sec y) = c.$ (Ans)

* Solve: $\frac{x\,dx + y\,dy}{x\,dy - y\,dx} = \sqrt{\left\{\frac{a^2-x^2-y^2}{x^2+y^2}\right\}}$

Sol^n: Here we change to polar co-ordinates by putting,

$x = r\cos\theta,\quad y = r\sin\theta$

$r^2 = x^2+y^2$

$2r\,dr = 2x\,dx + 2y\,dy$

$\therefore r\,dr = x\,dx + y\,dy$

$\frac{y}{x} = \tan\theta$

$\frac{x\,dy - y\,dx}{x^2} = \sec^2\theta\,d\theta$

$\Rightarrow x\,dy - y\,dx = x^2\sec^2\theta\,d\theta$

$\Rightarrow x\,dy - y\,dx = r^2\,d\theta$

the eq^n become

$\frac{r\,dr}{r^2\,d\theta} = \sqrt{\frac{a^2-r^2}{r^2}}$

$\Rightarrow \frac{1}{r}\frac{dr}{d\theta} = \sqrt{\frac{a^2-r^2}{r^2}}$

$\Rightarrow \frac{dr}{\sqrt{a^2-r^2}} = d\theta$

$\frac{r}{a} = \sin^{-1}(\theta + c)$

$\therefore \sqrt{x^2+y^2} = a\sin^{-1}\left(\tan^{-1}\frac{y}{x} + c\right)$

(Ans)

(viii) $x\frac{dy}{dx} - y = x\sqrt{x^2+y^2}\,dx$ —— (1)

Sol^n: let, $x = r\cos\theta$

$y = r\sin\theta$

$r^2 = x^2+y^2$

$\frac{y}{x} = \tan\theta$

$\Rightarrow \frac{x\,dy - y\,dx}{x^2} = \sec^2\theta\, d\theta$

$\Rightarrow x\,dy - y\,dx = x^2\sec^2\theta\, d\theta$

From, eq^n (1)

$x\,dy - y\,dx = x\sqrt{x^2+y^2}\,dx$

$\Rightarrow x^2\sec^2\theta\, d\theta = x r\, dx$

$\Rightarrow x\sec^2\theta\, d\theta = r\, dx$

$\Rightarrow r\cos\theta \cdot \frac{1}{\cos^2\theta}\, d\theta = r\, dx$

$\Rightarrow \sec\theta\, d\theta = dx$

$\Rightarrow \ln(\sec\theta + \tan\theta) = x + \ln c$

$\Rightarrow \ln\left(\frac{\sqrt{x^2+y^2}}{x} + \frac{y}{x}\right) = x + c.$

(Ans)

* $\frac{dy}{dx} + 1 = e^{x+y}$

Sol^n: let, $x+y = z$

$1 + \frac{dy}{dx} = \frac{dz}{dx}$

$\Rightarrow \frac{dz}{dx} = e^z$

$\Rightarrow e^{-z}\,dz = dx$

$\Rightarrow -e^{z} = x + c.$ [by integrating]

$\Rightarrow -e^{x+y} = x + c.$

(Ans)

* $\frac{dy}{dx} = \sqrt{y-x}$

Sol^n: let, $y - x = z^2$

$\frac{dy}{dx} - 1 = 2z\frac{dz}{dx}$

$\Rightarrow 1 + 2z\frac{dz}{dx} = \frac{dy}{dx}$

$\Rightarrow 1 + 2z\frac{dz}{dx} = z$

$\Rightarrow 2z\frac{dz}{dx} = z - 1$

$\Rightarrow \frac{2z}{z-1}\,dz = dx$

$\Rightarrow 2\cdot\frac{(z-1)+1}{z-1}\,dz = dx$

$\Rightarrow 2\,dz + \frac{2}{z-1}\,dz = dx$

$\Rightarrow 2z + 2\log|z-1| = x$

$\Rightarrow \sqrt{y-x} + \log|\sqrt{y-x} - 1| = \frac{x}{2} + c.$

(Ans)

(9) $\dfrac{x+yy_1}{xy_1-y} = \sqrt{\dfrac{1-x^2-y^2}{x^2+y^2}}$

$x = r\cos\theta \,,\quad y = r\sin\theta$

$r^2 = \cancel{\sqrt{\;}}\,x^2+y^2 \,,\quad \theta = \tan^{-1}\frac{y}{x}$

$2r\,dr = 2x\,dx + 2y\,dy \qquad \tan\theta = \frac{y}{x}$

$\therefore r\,dr = x\,dx + y\,dy \;—\; (\text{i})$

Again, $\tan\theta = \frac{y}{x}$

$\sec^2\theta\, d\theta = \dfrac{x\,dy - y\,dx}{x^2}$

$x\,dy - y\,dx = x^2 \sec^2\theta\, d\theta$

$x\,dy - y\,dx = r^2\, d\theta \;—\; (\text{ii})$

$(\text{i}) \div (\text{ii})$

$\dfrac{1}{r}\cdot\dfrac{dr}{d\theta} = \dfrac{x\,dx + y\,dy}{x\,dy - y\,dx}$

$\Rightarrow \dfrac{1}{r}\cdot\dfrac{dr}{d\theta} = \sqrt{\dfrac{1-r^2}{r^2}}$

$\dfrac{dr}{\sqrt{1-r^2}} = d\theta$

$\Rightarrow \sin^{-1} r = \theta + c$

$\Rightarrow \sin^{-1}\left(\sqrt{x^2+y^2}\right) = \left(\tan^{-1}\frac{y}{x}\right) + c$

$\therefore \sqrt{x^2+y^2} = \sin\left(\tan^{-1} y/x + c\right)$. (Ans)

10. $\dfrac{dy}{dx} = e^{ax+by}$

let, $ax+by = z$

$a + b\cdot\dfrac{dy}{dx} = \cancel{\dfrac{dy}{dz}}\;\dfrac{dz}{dx}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{1}{b}\left(\dfrac{dz}{dx} - a\right)$

$\Rightarrow \dfrac{1}{b}\left(\dfrac{dz}{dx} - a\right) = e^z$

$\Rightarrow \dfrac{dz}{dx} = be^z + a$

$\Rightarrow \dfrac{dz}{a+be^z} = dx$

$\Rightarrow \dfrac{-ae^{-z}}{b+ae^{-z}}\,dz = \;\; a\,dx$

$\log\left(b+ae^{-z}\right) = -ax + c$

$\therefore b + ae^{-z} = e^{-ax+c}$.

(Ans)